Representation learning-based association mapping

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Introduction

- Complex genetic diseases, including mental disorders **prevalent in 1 in 8 individuals worldwide**, pose significant challenges to global health despite concerted research efforts.
- Challenges such as missing heritability, polygenic traits, and high data dimensionality persist in genome-wide association studies (GWAS).
- To overcome these limitations, the study leverages **RL** weights in a "kernel" framework for association mapping using a Variational autoencoder (VAE). A weighted kernel using the SNVs and their reconstruction weights is formed:

$$K_w = \frac{1}{n}(G)^T W_r\, G$$

- G is a genotype matrix
- W = diag $(\widehat{\beta_1}, \ldots, \widehat{\beta_n})$, where n is the number of selected variants
- W_r is the reconstruction weights vector from the decoder and encoder layer.

$$\text{SKAT: } Q = y^T K_w\, y$$

- y is the vector of phenotype values in the GWAS dataset,
- K_w was calculated in the previous
- Q follows a mixture of the chi-squared distribution under the null hypothesis.

- The Q-score test, determines whether SNVs contribute to the observed phenotypic variation.

Method

RL-BAM: Representation Learning-Bridged Association Mapping

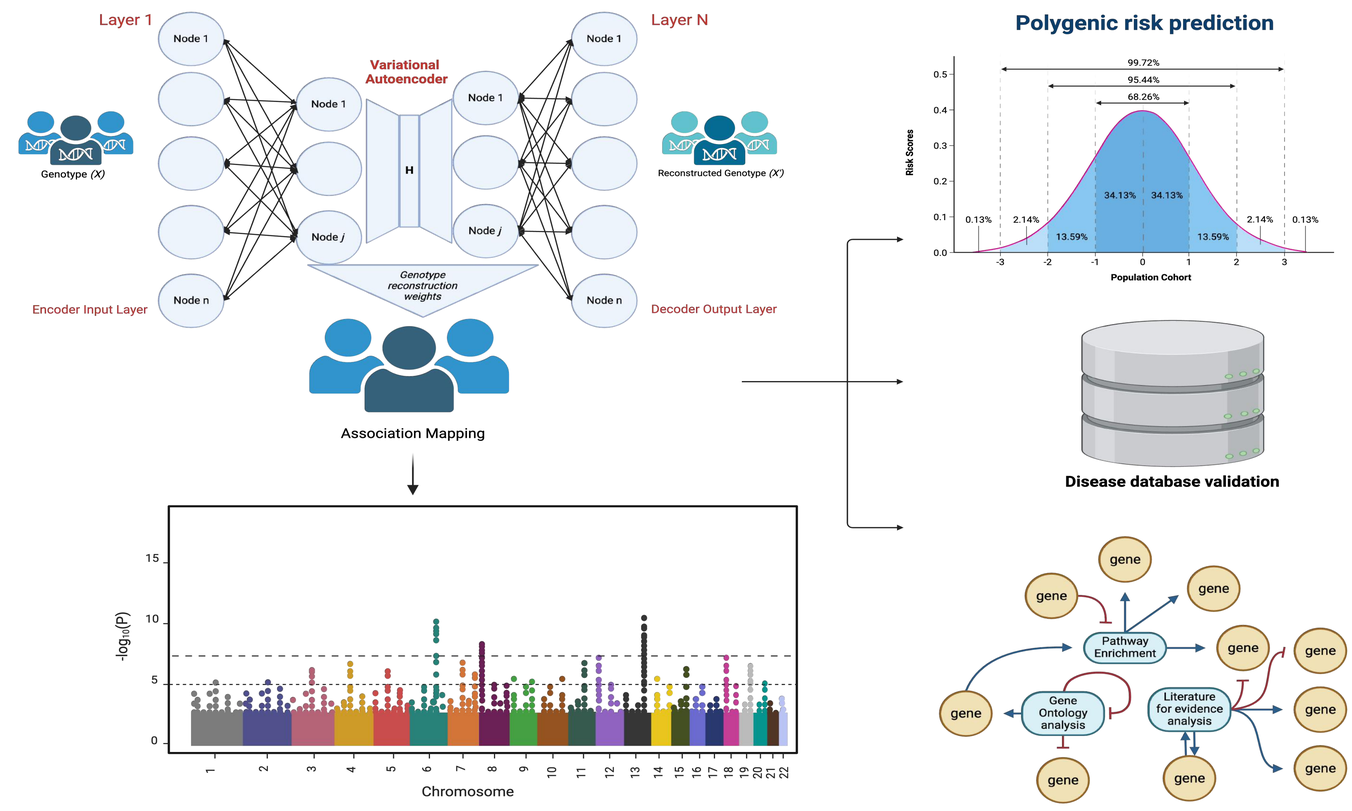
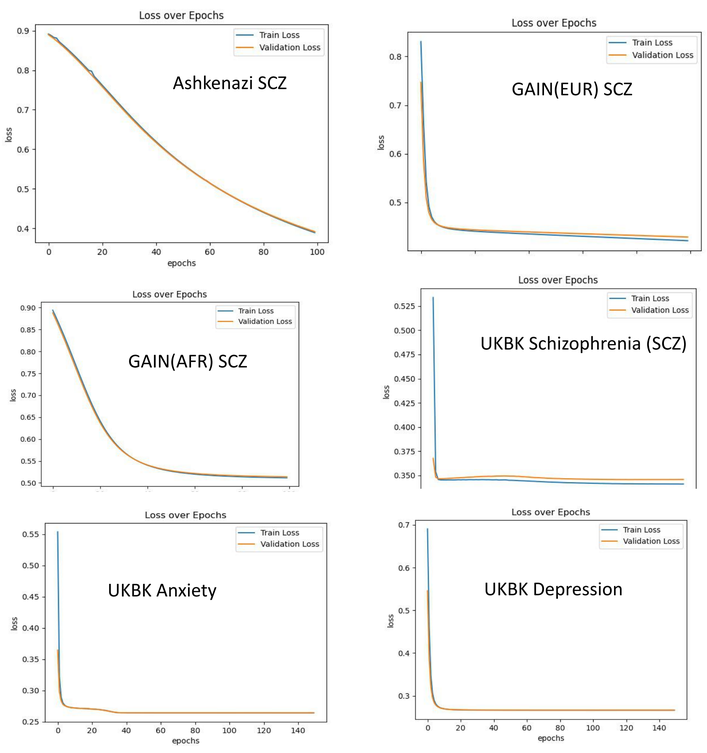


Figure 1: Representation Learning Based Association Mapping (RBAM) Developed using the Variational Autoencoder to reconstruct the input genotype. The reconstruction weights are then used in a weighted kernel for association mapping. Consequently, the association mapping results are represented by Manhattan plots, pathway enrichment, gene ontology analysis, disease database validation, and, eventually, polygenic risk predictions.

Results



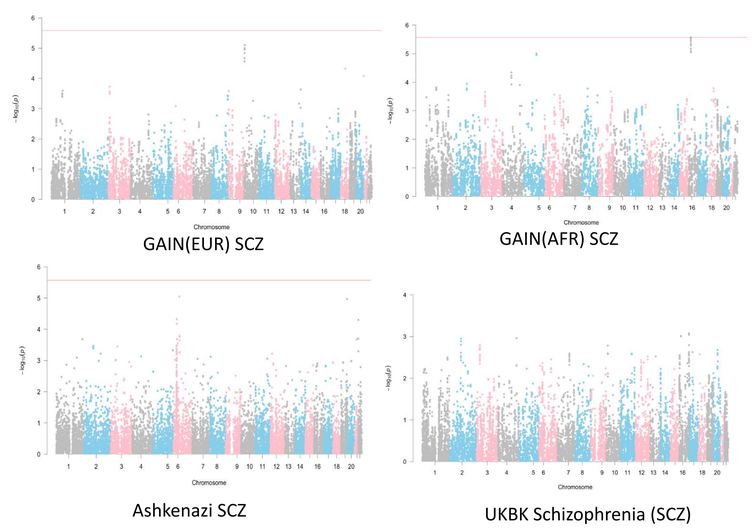
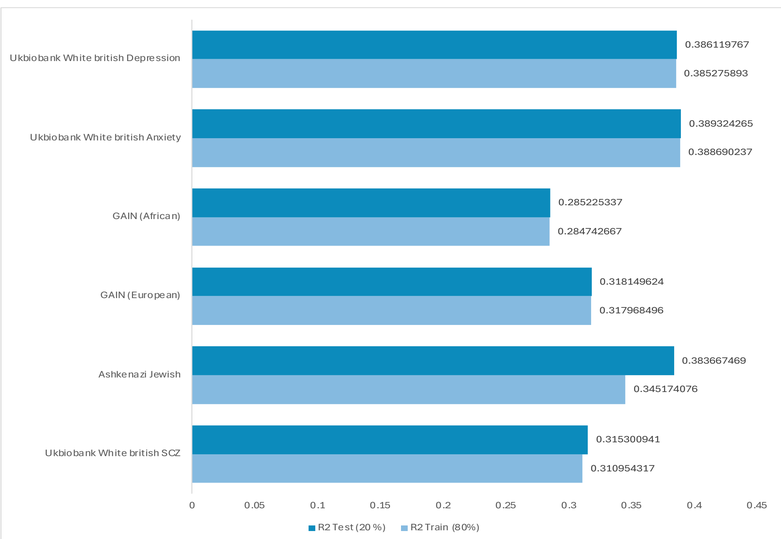


Figure 2 & 3 Loss (KL Divergence + Binary Cross Entropy) and Epochs curves & Manhattan plots of Bonferroni corrected association results by –log(p) values and chromosomes across multiple ancestries.



	MSE Train (80%)	MSE Test (20%)
UKBK Schizophrenia (SCZ)	0.17307885	0.1730789
Ashkenazi SCZ	0.60990703	0.6099171
GAIN(EUR) SCZ	0.29177266	0.2917719
GAIN(AFR) SCZ	0.29443702	0.2944374
UKBK Anxiety	0.13067332	0.13068548
UKBK Depression	0.13183363	0.13182738

Figure 6 & 7 R2 (Coefficient of determination) & MSE (Mean Squared Error) between the original and reconstructed genotype.

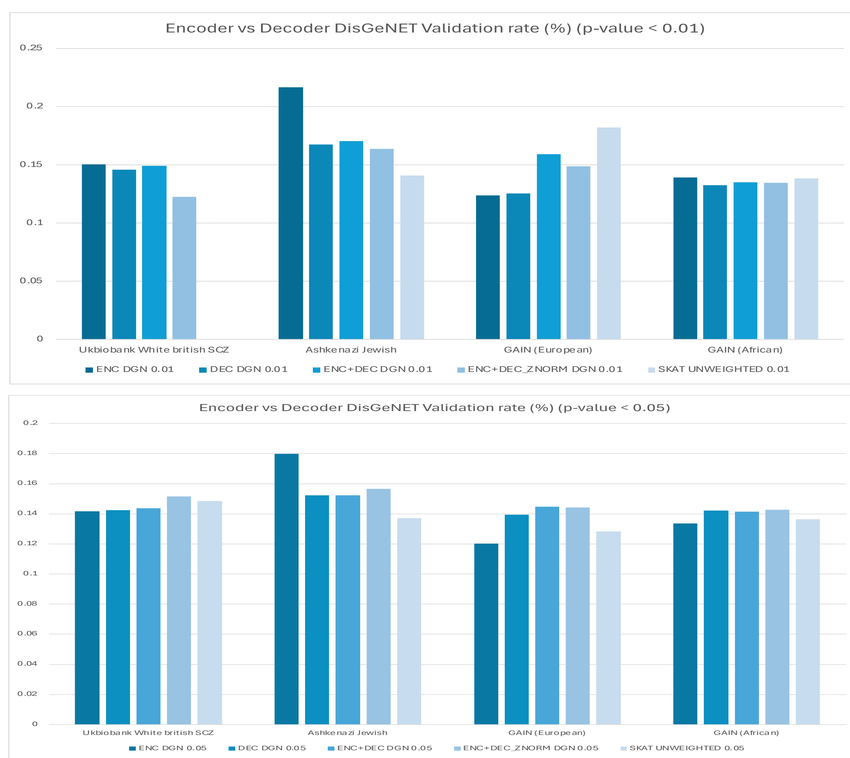


Figure 4 & 5 Disgenet Database validation at p-value < 0.01 &0.05 of encoder, decoder, encoder + decoder weighted kernel and Z-score normalized.

Conclusion

- Bayesian Hyperparameter optimization ensures the optimal performance of the VAE architecture with genotype reconstruction metrics, R2 & MSE, ranging from 0.28 to 0.39 and 0.13 to 0.61 across diverse ancestry cohorts and diseases.
- KEGG and Gene Ontology terms highlight potential biological mechanisms underlying Schizophrenia, including protein processing in the endoplasmic reticulum, cytoplasmic processes related to Parkinson's disease, ribosomal functions associated with membrane structures, and NF-kappa B signalling to regulate biological regulation.
- We show the potential of RL in advancing our understanding of complex genetic diseases and advancing precision medicine.

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